Kaj Hansteen Izora

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RESEARCH FOCUS

Robotics researcher working on uncertainty-aware embodied intelligence: perception, manipulation, and inference systems for robots that maintain calibrated beliefs, actively reduce epistemic uncertainty, and transfer reliably from simulation to the real world. Current emphasis on large-scale GPU simulation, simulation-based inference (neural posterior estimation, continuous normalizing flows, flow matching), and integrating these into real-robot autonomy stacks for manipulation.

RESEARCH INTERESTS

- Robot foundation models and the scaling/generalization properties of learned policies
- Manipulation and skill learning, in simulation and on real robot hardware
- Large-scale simulation and sim-to-real transfer (Isaac Sim, domain randomization, synthetic data)
- Uncertainty-aware learning: simulation-based inference (SBI), neural posterior estimation (NPE), normalizing flows, flow matching, diffusion
- Open, reproducible research — methods, datasets, and benchmarks shared with the community

EDUCATION

Portland State University

B.S. in Science, Data Science, and Social Science; Minor in Systems ScienceFall 2022 – Spring 2025

RESEARCH EXPERIENCE

NSF Research Training Group (RTG) Fellow — Robotics & Bayesian Learning

Portland State UniversityJun 2025 – Dec 2025

- Led the perception and inference stack for a robotic electric-vehicle (EV) battery disassembly platform, framing bolt-pose estimation as posterior inference under occlusion, sensor noise, and sim-to-real mismatch.
- Built a parallelized synthetic-data pipeline in **NVIDIA Isaac Sim on a multi-GPU cluster**, generating **1M+ domain-randomized samples** for uncertainty modeling and sim-to-real evaluation.
- Implemented Approximate Bayesian Computation (ABC) with PCA/UMAP embeddings as a baseline, then transitioned to **Sequential Neural Posterior Estimation (SNPE)** with learned summary networks for sample-efficient posterior approximation.
- Prototyped **continuous normalizing flows (CNFs), conditional flow matching, and image-conditioned posterior diffusion** from first principles on 2D toy posterior tasks; this exploration directly shaped the methodology section of a **DOE Genesis Mission grant proposal** that designates flow matching and CNFs as the lab's primary posterior-estimation direction.

Student Researcher — Teuscher Lab

Portland State UniversitySep 2024 – Jun 2025

- Designed a multi-agent simulation of Secret Hitler with a dual-layer prompt architecture separating private chain-of-thought reasoning from public speech, enabling measurement of the gap between agent intent and action.
- Analyzed emergent strategic adaptation, theory-of-mind reasoning, and implicit coordination across multi-agent LLM settings.
- Authored a first-author publication in the Northeast Journal of Complex Systems (2025, ~1,500 downloads); delivered an invited conference presentation.

APPLIED RESEARCH & ENGINEERING

Associate Physical AI Engineer — Rose City Robotics

Portland, ORDec 2025 – Present

- Real-robot manipulation: **Contribute to a Unitree G1 humanoid autonomy stack integrating perception, SLAM, reinforcement-learning locomotion, inverse-kinematics manipulation, and Kalman-fused localization; one demonstration application reached** 97% task success across 500+ trials**.**

- Sim-to-real perception: **Lead perception and synthetic-data work for an industrial defect-inspection program; fine-tuned a Dinomaly-based visual anomaly-detection model to** 100% image-level classification and localization **on 20 held-out real parts containing 64 sub-millimeter defects.**
- Synthetic-data pipeline: **Built a Blender CAD-to-synthetic-data pipeline with procedural defect generation and shader iteration via the Blender Model Context Protocol (MCP), producing photorealistic training data robust to lighting and pose variation.**

AI/Machine Learning Intern — The Contingent

Portland, ORJun 2024 – Sep 2024

- Built an AI pipeline using custom ML models and foundation models (Gemini, OpenAI, BERT) to automate student-to-internship matching; documented a 90% reduction in match placement time.

TEACHING & OUTREACH

Tech and Maker Space Associate — Self Enhancement, Inc.

Portland, ORSep 2024 – Jun 2025

- Designed and delivered an introductory AI and Python curriculum for Black youth with no prior coding experience, building accessible pathways into computer science and AI research.

PUBLICATIONS

Izora, K. H. *Exploring the Potential of Large Language Models (LLMs) to Simulate Social Group Dynamics.* **Northeast Journal of Complex Systems** (2025). First author; invited conference presentation. ~1,500 downloads.
orb.binghamton.edu/nejcs/vol7/iss2/5/

AWARDS & HONORS

- Portland State University Diversity Scholar (2022–2025)
- Portland State University President’s List (Fall 2021 – Winter 2025)
- Statistical Society NextGen Scholar (2023)

TECHNICAL SKILLS

Languages: Python (PyTorch, JAX, TensorFlow), C++, R, SQL, Bash

Robotics & Simulation: NVIDIA Isaac Sim, ROS2, SLAM, Unitree G1 humanoid, Blender, domain randomization, sim-to-real pipelines

ML Methods: Simulation-based inference, neural posterior estimation, normalizing flows, flow matching, diffusion models, representation learning, reinforcement learning, foundation-model fine-tuning

Infrastructure: Linux/Unix, Docker, HPC (Slurm), multi-GPU training, MCAP / DDS

SELECTED INDEPENDENT PROJECTS

- Continuous normalizing flows from scratch **— hand-built vector fields, RK4 ODE integrators, augmented state + log-density integration, maximum-likelihood CNF training on 2D toy posteriors.**
- Conditional flow matching **— target velocity fields derived from chosen probability paths, conditional posterior sampling on 2D transport tasks.**
- Image-conditioned posterior diffusion **— convolutional image encoder + noise-prediction denoiser on a partial-observation toy inverse problem, demonstrating multimodal-posterior behavior that deterministic regressors collapse.**
- Transformer / attention from first principles **— QKV implementation, patch-embedding loader, attention-map visualization for educational/teaching use.**